1.53 3D Vectors - The Vector Product

Worked Solutions

1 a)

$\mathbf{i} \times \mathbf{i} = \mathbf{0}$.

1 b)

$\mathbf{i} \times \mathbf{j} = \mathbf{k}$.

1 c)

$\mathbf{i} \times \mathbf{k} = -\mathbf{j}$.

1 d)

$\mathbf{j} \times \mathbf{i} = -\mathbf{k}$.

1 e)

$\mathbf{j} \times \mathbf{j} = \mathbf{0}$.

1 f)

$\mathbf{j} \times \mathbf{k} = \mathbf{i}$.

1 g)

$\mathbf{k} \times \mathbf{i} = \mathbf{j}$.

1 h)

$\mathbf{k} \times \mathbf{j} = -\mathbf{i}$.

1 i)

$\mathbf{k} \times \mathbf{k} = \mathbf{0}$.

2 a)

$\mathbf{i} \times (\mathbf{j} + \mathbf{k}) = \mathbf{i} \times \mathbf{j} + \mathbf{i} \times \mathbf{k} = \mathbf{k} - \mathbf{j}$.

2 b)

$\mathbf{j} \times (\mathbf{i} + \mathbf{k}) = \mathbf{j} \times \mathbf{i} + \mathbf{j} \times \mathbf{k} = -\mathbf{k} + \mathbf{i}$.

2 c)

$\mathbf{k} \times (\mathbf{i} + \mathbf{j}) = \mathbf{k} \times \mathbf{i} + \mathbf{k} \times \mathbf{j} = \mathbf{j} - \mathbf{i}$.

2 d)

$(\mathbf{i} - \mathbf{j}) \times (\mathbf{j} - \mathbf{k}) = \mathbf{i} \times \mathbf{j} - \mathbf{i} \times \mathbf{k} - \mathbf{j} \times \mathbf{j} + \mathbf{j} \times \mathbf{k} = \mathbf{k} + \mathbf{j} - \mathbf{0} + \mathbf{i} = \mathbf{i} + \mathbf{j} + \mathbf{k}$.

2 e)

$(\mathbf{k}-\mathbf{i})\times(\mathbf{j}-\mathbf{i}) = \mathbf{k}\times\mathbf{j}-\mathbf{k}\times\mathbf{i}-\mathbf{i}\times\mathbf{j}+\mathbf{i}\times\mathbf{i} = -\mathbf{i}-\mathbf{j}-\mathbf{k}+\mathbf{0} = -\mathbf{i}-\mathbf{j}-\mathbf{k}.$

2 f)

$$\begin{aligned}&(\mathbf{i}+\mathbf{j}+\mathbf{k})\times(-\mathbf{i}-\mathbf{j}+\mathbf{k})\\ &= -\mathbf{i}\times\mathbf{i}-\mathbf{i}\times\mathbf{j}+\mathbf{i}\times\mathbf{k}-\mathbf{j}\times\mathbf{i}-\mathbf{j}\times\mathbf{j}+\mathbf{j}\times\mathbf{k}-\mathbf{k}\times\mathbf{i}-\mathbf{k}\times\mathbf{j}+\mathbf{k}\times\mathbf{k}\\ &= \mathbf{0}-\mathbf{k}-\mathbf{j}+\mathbf{k}+\mathbf{0}+\mathbf{i}-\mathbf{j}+\mathbf{i}+\mathbf{0} = 2\mathbf{i}-2\mathbf{j}.\end{aligned}$$

2 g)

$$\begin{aligned}&(\mathbf{k}-\mathbf{j}-\mathbf{i})\times(\mathbf{j}-\mathbf{i}-\mathbf{k})\\ &= \mathbf{k}\times\mathbf{j}-\mathbf{k}\times\mathbf{i}-\mathbf{k}\times\mathbf{k}-\mathbf{j}\times\mathbf{j}+\mathbf{j}\times\mathbf{i}+\mathbf{j}\times\mathbf{k}-\mathbf{i}\times\mathbf{j}+\mathbf{i}\times\mathbf{i}+\mathbf{i}\times\mathbf{k}\\ &= -\mathbf{i}-\mathbf{j}-\mathbf{0}-\mathbf{0}-\mathbf{k}+\mathbf{i}-\mathbf{k}+\mathbf{0}-\mathbf{j} = -2\mathbf{j}-2\mathbf{k}.\end{aligned}$$

2 h)

$$\begin{aligned}&(2\mathbf{i}-3\mathbf{j}+5\mathbf{k})\times(3\mathbf{i}+\mathbf{j}-\mathbf{k})\\ &= 2\mathbf{i}\times3\mathbf{i}+2\mathbf{i}\times\mathbf{j}-2\mathbf{i}\times\mathbf{k}-3\mathbf{j}\times3\mathbf{i}-3\mathbf{j}\times\mathbf{j}+3\mathbf{j}\times\mathbf{k}+5\mathbf{k}\times3\mathbf{i}+5\mathbf{k}\times\mathbf{j}-5\mathbf{k}\times\mathbf{k}\\ &= \mathbf{0}+2\mathbf{k}+2\mathbf{j}+9\mathbf{k}-\mathbf{0}+3\mathbf{i}+15\mathbf{j}-5\mathbf{i}-\mathbf{0}\\ &= -2\mathbf{i}+17\mathbf{j}+11\mathbf{k}.\end{aligned}$$

2 i)

$$\begin{aligned}(3\mathbf{j}-5\mathbf{i}-3\mathbf{k})\times(6\mathbf{i}-8\mathbf{k}) &= 18\mathbf{j}\times\mathbf{i}-24\mathbf{j}\times\mathbf{k}-30\mathbf{i}\times\mathbf{i}+40\mathbf{i}\times\mathbf{k}-18\mathbf{k}\times\mathbf{i}+24\mathbf{k}\times\mathbf{k}\\ &= -18\mathbf{k}-24\mathbf{i}-\mathbf{0}-40\mathbf{j}-18\mathbf{j}+\mathbf{0} = -24\mathbf{i}-58\mathbf{j}-18\mathbf{k}.\end{aligned}$$

3 a)

$$\begin{pmatrix}1\\4\\2\end{pmatrix}\cdot\begin{pmatrix}8\\-5\\6\end{pmatrix} = 1\times8+4\times(-5)+2\times6 = 0$$

$$\begin{pmatrix}3\\6\\1\end{pmatrix}\cdot\begin{pmatrix}8\\-5\\6\end{pmatrix} = 3\times8+6\times(-5)+1\times6 = 0$$

Therefore $\begin{pmatrix}8\\-5\\6\end{pmatrix}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

3 b)

$$\begin{pmatrix}1\\4\\2\end{pmatrix} \times \begin{pmatrix}3\\6\\1\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\1 & 4 & 2\\3 & 6 & 1\end{vmatrix}$$
$$= (4 \times 1 - 2 \times 6)\mathbf{i} - (1 \times 1 - 2 \times 3)\mathbf{j} + (1 \times 6 - 4 \times 3)\mathbf{k}$$
$$= -8\mathbf{i} + 5\mathbf{j} - 6\mathbf{k} = \begin{pmatrix}-8\\5\\-6\end{pmatrix}.$$

4 a)

$$\begin{pmatrix}2\\2\\3\end{pmatrix} \times \begin{pmatrix}3\\2\\2\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\2 & 2 & 3\\3 & 2 & 2\end{vmatrix}$$
$$= (2 \times 2 - 3 \times 2)\mathbf{i} - (2 \times 2 - 3 \times 3)\mathbf{j} + (2 \times 2 - 2 \times 3)\mathbf{k}$$
$$= -2\mathbf{i} + 5\mathbf{j} - 2\mathbf{k} = \begin{pmatrix}-2\\5\\-2\end{pmatrix}.$$

4 b)

$\mathbf{a}$ and $\mathbf{b}$ are both perpendicular to $\mathbf{a} \times \mathbf{b}$.

5 a)

$$\begin{pmatrix}1\\2\\1\end{pmatrix} \times \begin{pmatrix}3\\1\\2\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\1 & 2 & 1\\3 & 1 & 2\end{vmatrix}$$
$$= (2 \times 2 - 1 \times 1)\mathbf{i} - (1 \times 2 - 1 \times 3)\mathbf{j} + (1 \times 1 - 2 \times 3)\mathbf{k}$$
$$= 3\mathbf{i} + \mathbf{j} - 5\mathbf{k}$$
$$= \begin{pmatrix}3\\1\\-5\end{pmatrix}.$$

5 b)

$$\begin{pmatrix}5\\3\\2\end{pmatrix} \times \begin{pmatrix}4\\1\\6\end{pmatrix} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\5 & 3 & 2\\4 & 1 & 6\end{vmatrix}$$
$$= (3 \times 6 - 2 \times 1)\mathbf{i} - (5 \times 6 - 2 \times 4)\mathbf{j} + (5 \times 1 - 3 \times 4)\mathbf{k}$$
$$= 16\mathbf{i} - 22\mathbf{j} - 7\mathbf{k}$$
$$= \begin{pmatrix}16\\-22\\-7\end{pmatrix}.$$

5 c)

$$\begin{pmatrix}-2\\3\\-9\end{pmatrix}\times\begin{pmatrix}-10\\0\\7\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-2&3&-9\\-10&0&7\end{vmatrix}$$
$$=(3\times 7-(-9)\times 0)\mathbf{i}-((-2)\times 7-(-9)\times(-10))\mathbf{j}+((-2)\times 0-3\times(-10))\mathbf{k}$$
$$=21\mathbf{i}+104\mathbf{j}+30\mathbf{k}$$
$$=\begin{pmatrix}21\\104\\30\end{pmatrix}.$$

5 d)

$$\begin{pmatrix}3\\8\\-2\end{pmatrix}\times\begin{pmatrix}5\\-1\\1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&8&-2\\5&-1&1\end{vmatrix}$$
$$=(8\times 1-(-2)\times(-1))\mathbf{i}-(3\times 1-(-2)\times 5)\mathbf{j}+(3\times(-1)-8\times 5)\mathbf{k}$$
$$=6\mathbf{i}-13\mathbf{j}-43\mathbf{k}.$$

5 e)

$$\begin{pmatrix}-10\\2\\-3\end{pmatrix}\times\begin{pmatrix}-1\\5\\4\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-10&2&-3\\-1&5&4\end{vmatrix}$$
$$=(2\times 4-(-3)\times 5)\mathbf{i}-((-10)\times 4-(-3)\times(-1))\mathbf{j}+((-10)\times 5-2\times(-1))\mathbf{k}$$
$$=23\mathbf{i}+43\mathbf{j}-48\mathbf{k}.$$

5 f)

$$\begin{pmatrix}3\\0\\7\end{pmatrix}\times\begin{pmatrix}0\\-2\\-1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&0&7\\0&-2&-1\end{vmatrix}$$
$$=(0\times(-1)-7\times(-2))\mathbf{i}-(3\times(-1)-7\times 0)\mathbf{j}+(3\times(-2)-0\times 0)\mathbf{k}$$
$$=14\mathbf{i}+3\mathbf{j}-6\mathbf{k}.$$

5 g)

$$\begin{pmatrix}5\\0\\0\end{pmatrix}\times\begin{pmatrix}2\\-3\\1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\5&0&0\\2&-3&1\end{vmatrix}$$
$$=(0\times 1-0\times(-3))\mathbf{i}-(5\times 1-0\times 2)\mathbf{j}+(5\times(-3)-0\times 2)\mathbf{k}$$
$$=-5\mathbf{j}-15\mathbf{k}.$$

5 h)

$$\begin{pmatrix}-3\\0\\10\end{pmatrix}\times\begin{pmatrix}0\\3\\0\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-3&0&10\\0&3&0\end{vmatrix}$$
$$=(0\times 0-10\times 3)\mathbf{i}-((-3)\times 0-10\times 0)\mathbf{j}+((-3)\times 3-0\times 0)\mathbf{k}$$
$$=-30\mathbf{i}-9\mathbf{k}.$$

6)

$$\begin{pmatrix}2\\0\\-3\end{pmatrix}\times\begin{pmatrix}0\\-3\\-1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&0&-3\\0&-3&-1\end{vmatrix}$$
$$=-9\mathbf{i}+2\mathbf{j}-6\mathbf{k}.$$
$$|-9\mathbf{i}+2\mathbf{j}-6\mathbf{k}|=\sqrt{(-9)^2+2^2+(-6)^2}=\sqrt{121}=11.$$

So the unit vector is $\frac{1}{11}(-9\mathbf{i}+2\mathbf{j}-6\mathbf{k})=-\frac{9}{11}\mathbf{i}+\frac{2}{11}\mathbf{j}-\frac{6}{11}\mathbf{k}$, or $\begin{pmatrix}-9/11\\2/11\\-6/11\end{pmatrix}$.

7)

$$\begin{pmatrix}-1\\1\\2\end{pmatrix}\times\begin{pmatrix}5\\-2\\0\end{pmatrix}=4\mathbf{i}+10\mathbf{j}-3\mathbf{k}.$$
$$|4\mathbf{i}+10\mathbf{j}-3\mathbf{k}|=\sqrt{4^2+10^2+(-3)^2}=\sqrt{125}=5\sqrt{5}.$$

So the unit vector is $\frac{1}{5\sqrt{5}}(4\mathbf{i}+10\mathbf{j}-3\mathbf{k})=\frac{4\sqrt{5}}{25}\mathbf{i}+\frac{2\sqrt{5}}{5}\mathbf{j}-\frac{3\sqrt{5}}{25}\mathbf{k}$, or $\begin{pmatrix}4\sqrt{5}/25\\2\sqrt{5}/5\\-3\sqrt{5}/25\end{pmatrix}$.

8)

$$\begin{pmatrix}1\\a\\2\end{pmatrix}\times\begin{pmatrix}a\\2\\3\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&a&2\\a&2&3\end{vmatrix}$$
$$=(3a-4)\mathbf{i}+(2a-3)\mathbf{j}+(2-a^2)\mathbf{k}$$
$$=\begin{pmatrix}3a-4\\2a-3\\2-a^2\end{pmatrix}.$$

9)

$$\begin{pmatrix} p \\ 3 \\ -2 \end{pmatrix} \times \begin{pmatrix} 1-p \\ 2 \\ p \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ p & 3 & -2 \\ 1-p & 2 & p \end{vmatrix}$$
$$= (3p+4)\mathbf{i} + (-p^2+2p-2)\mathbf{j} + (5p-3)\mathbf{k}$$
$$= \begin{pmatrix} 3p+4 \\ -p^2+2p-2 \\ 5p-3 \end{pmatrix}.$$

10)

$$\begin{pmatrix} 1 \\ q \\ 1-q \end{pmatrix} \times \begin{pmatrix} q-1 \\ 1 \\ q \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & q & 1-q \\ q-1 & 1 & q \end{vmatrix}$$
$$= (q^2+q-1)\mathbf{i} + (-q^2+q-1)\mathbf{j} + (-q^2+q+1)\mathbf{k}$$
$$= \begin{pmatrix} q^2+q-1 \\ -q^2+q-1 \\ -q^2+q+1 \end{pmatrix}.$$

11 a)

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 2 \\ -7 \\ 3 \end{pmatrix} \times \begin{pmatrix} 4 \\ -8 \\ -11 \end{pmatrix}$$
$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -7 & 3 \\ 4 & -8 & -11 \end{vmatrix}$$
$$= 101\mathbf{i} + 34\mathbf{j} + 12\mathbf{k}$$
$$= \begin{pmatrix} 101 \\ 34 \\ 12 \end{pmatrix}.$$

11 b)

$$\mathbf{b} \times \mathbf{a} = \begin{pmatrix} 4 \\ -8 \\ -11 \end{pmatrix} \times \begin{pmatrix} 2 \\ -7 \\ 3 \end{pmatrix}$$
$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -8 & -11 \\ 2 & -7 & 3 \end{vmatrix}$$
$$= -101\mathbf{i} - 34\mathbf{j} - 12\mathbf{k}$$
$$= \begin{pmatrix} -101 \\ -34 \\ -12 \end{pmatrix}.$$

11 c)

As $\mathbf{a} \times \mathbf{b} = 101\mathbf{i} + 34\mathbf{j} + 12\mathbf{k}$ and $\mathbf{b} \times \mathbf{a} = -101\mathbf{i} - 34\mathbf{j} - 12\mathbf{k}$, we have that $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$.

12)

$$\begin{aligned}\begin{pmatrix} 3 \\ -6 \\ 12 \end{pmatrix} \times \begin{pmatrix} -4 \\ 8 \\ -16 \end{pmatrix} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -6 & 12 \\ -4 & 8 & -16 \end{vmatrix} \\ &= 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} \\ &= \mathbf{0}.\end{aligned}$$

Therefore the vectors are parallel.

13)

$$\begin{aligned}\text{Area of triangle} &= \frac{1}{2}|\overrightarrow{OA} \times \overrightarrow{OB}| \\ &= \frac{1}{2}|\sqrt{2}\mathbf{i} - 9\mathbf{j} - \sqrt{5}\mathbf{k}| \\ = \frac{1}{2}\sqrt{(\sqrt{2})^2 + (-9)^2 + (-\sqrt{5})^2} &= \frac{1}{2}\sqrt{2 + 81 + 5} = \frac{1}{2}\sqrt{88} = \sqrt{22} \text{ units}^2.\end{aligned}$$

14)

$$\begin{aligned}\begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix} \times \begin{pmatrix} 8 \\ 3 \\ 5 \end{pmatrix} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -2 & 5 \\ 8 & 3 & 5 \end{vmatrix} \\ &= -25\mathbf{i} + 25\mathbf{j} + 25\mathbf{k}.\end{aligned}$$

Area of parallelogram $= |-25\mathbf{i} + 25\mathbf{j} + 25\mathbf{k}| = \sqrt{(-25)^2 + 25^2 + 25^2} = \sqrt{1875} = 25\sqrt{3}$ units^2.

15 a)

$$\begin{aligned}\overrightarrow{OA} &= \overrightarrow{BC} \\ &= \begin{pmatrix} 3 \\ 6 \\ 12 \end{pmatrix}, \qquad \overrightarrow{OB} \\ &= \overrightarrow{AC} \\ &= \begin{pmatrix} 5 \\ -10 \\ 20 \end{pmatrix}. \\ \overrightarrow{OC} &= \overrightarrow{OA} + \overrightarrow{AC} \\ &= \overrightarrow{OA} + \overrightarrow{OB} \\ &= \begin{pmatrix} 3 \\ 6 \\ 12 \end{pmatrix} + \begin{pmatrix} 5 \\ -10 \\ 20 \end{pmatrix} \\ &= \begin{pmatrix} 8 \\ -4 \\ 32 \end{pmatrix}.\end{aligned}$$

Therefore $C(8, -4, 32)$.

15 b)

$$\begin{pmatrix} 3 \\ 6 \\ 12 \end{pmatrix} \times \begin{pmatrix} 5 \\ -10 \\ 20 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 6 & 12 \\ 5 & -10 & 20 \end{vmatrix}$$
$$= 240\mathbf{i} + 0\mathbf{j} - 60\mathbf{k}.$$

Area of parallelogram $= |240\mathbf{i} - 60\mathbf{k}| = \sqrt{240^2 + (-60)^2} = \sqrt{61200} = 60\sqrt{17}$ units^2.

16)

$$\overrightarrow{OA} = \begin{pmatrix} 3 \\ -6 \\ -2 \end{pmatrix}, \quad \overrightarrow{OB}$$
$$= \begin{pmatrix} 4 \\ -6 \\ 0 \end{pmatrix}.$$

$$\begin{pmatrix} 3 \\ -6 \\ -2 \end{pmatrix} \times \begin{pmatrix} 4 \\ -6 \\ 0 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -6 & -2 \\ 4 & -6 & 0 \end{vmatrix}$$
$$= -12\mathbf{i} - 8\mathbf{j} + 6\mathbf{k}.$$

$$\text{Area of triangle} = \frac{1}{2}|-12\mathbf{i} - 8\mathbf{j} + 6\mathbf{k}|$$
$$= \frac{1}{2}\sqrt{(-12)^2 + (-8)^2 + 6^2}$$
$$= \frac{1}{2}\sqrt{244}$$
$$= \sqrt{61} \text{ units}^2.$$

17 a)

Let A, B, C be $(-1, 0, 1), (2, 1, 0), (1, 1, -1)$ respectively.

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$
$$= \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$
$$= \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}, \quad \overrightarrow{AC}$$
$$= \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} - \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$
$$= \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}.$$

$$\begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} \times \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = -\mathbf{i} - 4\mathbf{j} + \mathbf{k}.$$

$$\text{Area} = \frac{1}{2}\sqrt{(-1)^2 + (-4)^2 + 1^2} = \frac{1}{2}\sqrt{18} = \frac{3\sqrt{2}}{2} \text{ units}^2.$$

17 b)

Let A, B, C be $(3, -8, 1), (2, 1, -3), (1, 2, 3)$ respectively.

$$\begin{aligned} \overrightarrow{AB} &= \begin{pmatrix} -1 \\ 9 \\ -4 \end{pmatrix}, \quad \overrightarrow{AC} \\ &= \begin{pmatrix} -2 \\ 10 \\ 2 \end{pmatrix}, \quad \overrightarrow{AB} \times \overrightarrow{AC} \\ &= 58\mathbf{i} - 10\mathbf{j} + 8\mathbf{k}. \end{aligned}$$

$$\text{Area} = \frac{1}{2}\sqrt{58^2 + (-10)^2 + 8^2} = \frac{1}{2}\sqrt{3528} = 21\sqrt{2} \text{ units}^2.$$

17 c)

Let A, B, C be $(4, 2, -6), (3, 0, -1), (8, 3, -5)$ respectively.

$$\begin{aligned} \overrightarrow{AB} &= \begin{pmatrix} -1 \\ -2 \\ 5 \end{pmatrix}, \quad \overrightarrow{AC} \\ &= \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix}, \quad \overrightarrow{AB} \times \overrightarrow{AC} \\ &= -7\mathbf{i} + 21\mathbf{j} + 7\mathbf{k}. \end{aligned}$$

$$\text{Area} = \frac{1}{2}\sqrt{(-7)^2 + 21^2 + 7^2} = \frac{1}{2}\sqrt{539} = \frac{7}{2}\sqrt{11} \text{ units}^2.$$

17 d)

Let A, B, C be $(-1, 4, 5), (9, -2, -3), (0, 5, 5)$ respectively.

$$\begin{aligned} \overrightarrow{AB} &= \begin{pmatrix} 10 \\ -6 \\ -8 \end{pmatrix}, \quad \overrightarrow{AC} \\ &= \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \overrightarrow{AB} \times \overrightarrow{AC} \\ &= 8\mathbf{i} - 8\mathbf{j} + 16\mathbf{k}. \end{aligned}$$

$$\text{Area} = \frac{1}{2}\sqrt{8^2 + (-8)^2 + 16^2} = \frac{1}{2}\sqrt{384} = 4\sqrt{6} \text{ units}^2.$$

18 a)

$$\begin{aligned}\overrightarrow{OA} &= \begin{pmatrix}5\\2\\-1\end{pmatrix}, \quad \overrightarrow{OB} \\ &= \begin{pmatrix}9\\-4\\2\end{pmatrix}, \quad \overrightarrow{OC} \\ &= \begin{pmatrix}-17\\16\\-8\end{pmatrix}.\end{aligned}$$

$$p\begin{pmatrix}5\\2\\-1\end{pmatrix} + q\begin{pmatrix}9\\-4\\2\end{pmatrix} = \begin{pmatrix}-17\\16\\-8\end{pmatrix}$$

$$5p + 9q = -17 \quad (1), \qquad 2p - 4q = 16 \quad (2).$$

Therefore $p = 2, q = -3$. Check: $2(-1) + (-3)(2) = -8$. Therefore $2\overrightarrow{OA} - 3\overrightarrow{OB} = \overrightarrow{OC}$, so the vectors are linearly dependent.

18 b)

$$\begin{aligned}\overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \\ &= \begin{pmatrix}4\\-6\\3\end{pmatrix}, \quad \overrightarrow{AC} \\ &= \overrightarrow{OC} - \overrightarrow{OA} \\ &= \begin{pmatrix}-22\\14\\-7\end{pmatrix}.\end{aligned}$$

$$\begin{aligned}\overrightarrow{AB} \times \overrightarrow{AC} &= \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\4 & -6 & 3\\-22 & 14 & -7\end{vmatrix} \\ &= 0\mathbf{i} - 38\mathbf{j} - 76\mathbf{k}.\end{aligned}$$

$$\text{Area} = \frac{1}{2}\sqrt{(-38)^2 + (-76)^2} = \frac{1}{2}\sqrt{7220} = 19\sqrt{5} \text{ units}^2.$$

19)

$$\begin{aligned}\overrightarrow{OA} &= \begin{pmatrix}2\\-5\\6\end{pmatrix}, \quad \overrightarrow{OB} \\ &= \begin{pmatrix}4\\-3\\-1\end{pmatrix}, \quad \overrightarrow{OC} \\ &= \begin{pmatrix}-22\\6\\24\end{pmatrix}.\end{aligned}$$

$$p\begin{pmatrix}2\\-5\\6\end{pmatrix}+q\begin{pmatrix}4\\-3\\-1\end{pmatrix}=\begin{pmatrix}-22\\6\\24\end{pmatrix}$$

$$2p+4q=-22 \quad (1), \qquad -5p-3q=6 \quad (2).$$

Therefore $p=3, q=-7$. Check: $6(3)+(-1)(-7)=25\neq 24$. So $\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC}$ are linearly independent.

20 a)

$$\begin{pmatrix}2\\3\\-1\end{pmatrix}\times\begin{pmatrix}4\\2\\-3\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&3&-1\\4&2&-3\end{vmatrix}$$
$$=-7\mathbf{i}+2\mathbf{j}-8\mathbf{k}.$$

$$\begin{pmatrix}-7\\2\\-8\end{pmatrix}\cdot\begin{pmatrix}-2\\1\\2\end{pmatrix}=(-7)(-2)+2(1)+(-8)(2)=0.$$

Therefore the three vectors are coplanar.

20 b)

$$\begin{pmatrix}-4\\3\\6\end{pmatrix}\times\begin{pmatrix}-5\\3\\-8\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-4&3&6\\-5&3&-8\end{vmatrix}$$
$$=-42\mathbf{i}-62\mathbf{j}+3\mathbf{k}.$$

$$\begin{pmatrix}-42\\-62\\3\end{pmatrix}\cdot\begin{pmatrix}3\\-9\\-144\end{pmatrix}=(-42)(3)+(-62)(-9)+3(-144)=0.$$

Therefore the three vectors are coplanar.

21 a)

$$\mathbf{a}\times\mathbf{b}=\begin{pmatrix}-4\\8\\1\end{pmatrix}\times\begin{pmatrix}4\\-6\\3\end{pmatrix}$$
$$=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\-4&8&1\\4&-6&3\end{vmatrix}$$
$$=30\mathbf{i}+16\mathbf{j}-8\mathbf{k}$$
$$=\begin{pmatrix}30\\16\\-8\end{pmatrix}.$$

21 b)

$$\sin\theta = \frac{|\mathbf{a}\times\mathbf{b}|}{|\mathbf{a}||\mathbf{b}|} = \frac{|(30,16,-8)|}{|(-4,8,1)||(4,-6,3)|}$$

$$\Rightarrow \sin\theta = \frac{\sqrt{30^2+16^2+(-8)^2}}{\sqrt{(-4)^2+8^2+1^2}\sqrt{4^2+(-6)^2+3^2}}$$

$$= \frac{\sqrt{1220}}{\sqrt{81}\sqrt{61}}$$

$$= \frac{2\sqrt{305}}{9\sqrt{61}}$$

$$= \frac{2\sqrt{5}}{9}.$$

22 a)

$$\mathbf{a}\times\mathbf{b} = \begin{pmatrix}1\\0\\1\end{pmatrix}\times\begin{pmatrix}2\\-1\\0\end{pmatrix}$$

$$= \mathbf{i}+2\mathbf{j}-\mathbf{k}.$$

$$\sin\theta = \frac{|\mathbf{a}\times\mathbf{b}|}{|\mathbf{a}||\mathbf{b}|}$$

$$= \frac{\sqrt{1^2+2^2+(-1)^2}}{\sqrt{1^2+0^2+1^2}\sqrt{2^2+(-1)^2+0^2}}$$

$$= \frac{\sqrt{6}}{\sqrt{2}\sqrt{5}}$$

$$= \frac{\sqrt{15}}{5} \Rightarrow \theta$$

$$= 50.768\ldots^\circ.$$

Acute angle is 50.8° (1 DP).

22 b)

$$\mathbf{a}\times\mathbf{b} = \begin{pmatrix}-7\\12\\1\end{pmatrix}\times\begin{pmatrix}3\\-20\\-1\end{pmatrix}$$

$$= 8\mathbf{i}+4\mathbf{j}+104\mathbf{k}.$$

$$\sin\theta = \frac{\sqrt{8^2+4^2+104^2}}{\sqrt{(-7)^2+12^2+1^2}\sqrt{3^2+(-20)^2+(-1)^2}}$$

$$= \frac{\sqrt{10896}}{\sqrt{194}\sqrt{410}}$$

$$= \sqrt{\frac{10896}{79540}} \Rightarrow \theta$$

$$= 21.722\ldots^\circ.$$

Acute angle is 21.7° (1 DP).

22 c)

$$\begin{aligned}\mathbf{a}\times\mathbf{b} &= \begin{pmatrix}2\sqrt{2}\\0\\7\sqrt{2}\end{pmatrix}\times\begin{pmatrix}5\sqrt{2}\\2\sqrt{2}\\0\end{pmatrix}\\ &= -28\mathbf{i}+70\mathbf{j}+8\mathbf{k}.\end{aligned}$$

$$\begin{aligned}\sin\theta &= \frac{\sqrt{(-28)^2+70^2+8^2}}{\sqrt{(2\sqrt{2})^2+0^2+(7\sqrt{2})^2}\sqrt{(5\sqrt{2})^2+(2\sqrt{2})^2+0^2}}\\ &= \frac{\sqrt{5748}}{\sqrt{106}\sqrt{58}}\\ &= \sqrt{\frac{5748}{6148}}\Rightarrow\theta\\ &= 75.222\ldots^\circ.\end{aligned}$$

Acute angle is 75.2° (1 DP).

22 d)

$$\begin{aligned}\mathbf{a}\times\mathbf{b} &= \begin{pmatrix}1\\0\\9\end{pmatrix}\times\begin{pmatrix}0\\-1\\-15\end{pmatrix}\\ &= 9\mathbf{i}+15\mathbf{j}-\mathbf{k}.\end{aligned}$$

$$\begin{aligned}\sin\theta &= \frac{\sqrt{9^2+15^2+(-1)^2}}{\sqrt{1^2+0^2+9^2}\sqrt{0^2+(-1)^2+(-15)^2}}\\ &= \sqrt{\frac{307}{18532}}\Rightarrow\theta\\ &= 7.394\ldots^\circ.\end{aligned}$$

Acute angle is 7.4° (1 DP).

22 e)

$$\begin{aligned}\mathbf{a}\times\mathbf{b} &= \begin{pmatrix}\sqrt{2}\\\sqrt{3}\\0\end{pmatrix}\times\begin{pmatrix}\sqrt{3}\\\sqrt{2}\\\sqrt{6}\end{pmatrix}\\ &= 3\sqrt{2}\mathbf{i}-2\sqrt{3}\mathbf{j}-\mathbf{k}.\end{aligned}$$

$$\begin{aligned}\sin\theta &= \frac{\sqrt{(3\sqrt{2})^2+(-2\sqrt{3})^2+(-1)^2}}{\sqrt{(\sqrt{2})^2+(\sqrt{3})^2+0^2}\sqrt{(\sqrt{3})^2+(\sqrt{2})^2+(\sqrt{6})^2}}\\ &= \sqrt{\frac{31}{55}}\Rightarrow\theta\\ &= 48.656\ldots^\circ.\end{aligned}$$

Acute angle is 48.7° (1 DP).

22 f)

$$\begin{aligned}\mathbf{a}\times\mathbf{b} &= \begin{pmatrix}\pi\\-\sqrt{2}\\0\end{pmatrix}\times\begin{pmatrix}\sqrt{5}\\1\\\pi\end{pmatrix}\\ &= -\sqrt{2}\pi\mathbf{i}-\pi^2\mathbf{j}+(\pi+\sqrt{10})\mathbf{k}.\end{aligned}$$

$$\begin{aligned}\sin\theta &= \frac{\sqrt{(-\sqrt{2}\pi)^2+(-\pi^2)^2+(\pi+\sqrt{10})^2}}{\sqrt{\pi^2+(-\sqrt{2})^2+0^2}\sqrt{(\sqrt{5})^2+1^2+\pi^2}}\\ &= \frac{\sqrt{\pi^4+3\pi^2+2\sqrt{10}\pi+10}}{\sqrt{\pi^2+2}\sqrt{\pi^2+6}}\\ &= 0.912625\ldots\Rightarrow\theta\\ &= 65.870\ldots^\circ.\end{aligned}$$

Acute angle is 65.9° (1 DP).

23)

$$\begin{pmatrix}a\\a\\5\end{pmatrix}\times\begin{pmatrix}-1\\2\\-4\end{pmatrix}=\begin{pmatrix}-22\\7\\9\end{pmatrix}\Rightarrow\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\a&a&5\\-1&2&-4\end{vmatrix}$$

$$=-22\mathbf{i}+7\mathbf{j}+9\mathbf{k}$$

$$\Rightarrow(-4a-10)\mathbf{i}+(4a-5)\mathbf{j}+3a\mathbf{k}=-22\mathbf{i}+7\mathbf{j}+9\mathbf{k}.$$

$$-4a-10=-22\Rightarrow a=3,\qquad 4a-5=7\Rightarrow a=3,\qquad 3a=9\Rightarrow a=3.$$

24)

$$\begin{pmatrix}p\\3\\-2\end{pmatrix}\times\begin{pmatrix}4\\5\\2\end{pmatrix}=\begin{pmatrix}16\\q\\-2\end{pmatrix}\Rightarrow(16)\mathbf{i}+(-2p-8)\mathbf{j}+(5p-12)\mathbf{k}$$

$$=16\mathbf{i}+q\mathbf{j}-2\mathbf{k}.$$

$$-2p-8=q\quad(1),\qquad 5p-12=-2\Rightarrow p=2\quad(2).$$

Substitute (2) into (1): $-2(2)-8=q\Rightarrow q=-12$. Therefore $p=2$ and $q=-12$.

25)

$$\begin{pmatrix}8\\2m\\-9\end{pmatrix}\times\begin{pmatrix}5\\-3\\-1\end{pmatrix}=(-2m-27)\mathbf{i}-37\mathbf{j}+(-24-10m)\mathbf{k}$$

$$=-45\mathbf{i}-37\mathbf{j}+3n\mathbf{k}.$$

$$-2m-27=-45\Rightarrow m=9,\qquad -24-10\times 9=3n\Rightarrow n=-38.$$

26)

$$\begin{pmatrix} p \\ -3 \\ 8 \end{pmatrix} \times \begin{pmatrix} 10 \\ 5 \\ 2q \end{pmatrix} = (-6q - 40)\mathbf{i} - (2pq - 80)\mathbf{j} + (5p + 30)\mathbf{k}$$
$$= -16\mathbf{i} + 168\mathbf{j} + 5r\mathbf{k}.$$
$$-6q - 40 = -16 \Rightarrow q = -4, \quad -(2p \times (-4) - 80) = 168 \Rightarrow p = 11, \quad 5 \times 11 + 30 = 5r \Rightarrow r = 17.$$

27)

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ q-1 & q & -3 \\ 2 & 2q-3 & q \end{vmatrix} = (q^2 + 6q - 9)\mathbf{i} - (q^2 - q + 6)\mathbf{j} + (2q^2 - 7q + 3)\mathbf{k}.$$

If $\mathbf{a}$ is perpendicular to $\mathbf{b} \times \mathbf{c}$, then $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$.

$$2(q^2 + 6q - 9) + (-2)(-(q^2 - q + 6)) + (-1)(2q^2 - 7q + 3) = 0$$
$$\Rightarrow 2q^2 + 12q - 18 + 2q^2 - 2q + 12 - 2q^2 + 7q - 3 = 0 \Rightarrow 2q^2 + 17q - 9 = 0$$
$$\Rightarrow (2q - 1)(q + 9) = 0 \Rightarrow q = \frac{1}{2} \text{ or } q = -9.$$

28 a)

$(2\mathbf{a}) \times (5\mathbf{b}) = 10(\mathbf{a} \times \mathbf{b}) = 10\mathbf{c}$.

28 b)

$(\frac{1}{2}\mathbf{b}) \times (8\mathbf{a}) = 4(\mathbf{b} \times \mathbf{a}) = -4(\mathbf{a} \times \mathbf{b}) = -4\mathbf{c}$.

29)

If $\mathbf{a}$ is parallel to both $\mathbf{b}$ and $\mathbf{c}$ then $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ and $\mathbf{a} \times \mathbf{c} = \mathbf{0}$. Therefore $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c} = \mathbf{0} + \mathbf{0} = \mathbf{0}$.

30 a)

$$(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} - \mathbf{b}) = \mathbf{a} \times \mathbf{a} - \mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{a} - \mathbf{b} \times \mathbf{b}$$
$$= \mathbf{0} - \mathbf{a} \times \mathbf{b} - \mathbf{a} \times \mathbf{b} - \mathbf{0} = -2\mathbf{a} \times \mathbf{b}.$$

30 b)

$$(2\mathbf{a} - \mathbf{b}) \times (\mathbf{a} + 3\mathbf{b}) = 2\mathbf{a} \times \mathbf{a} + 6\mathbf{a} \times \mathbf{b} - \mathbf{b} \times \mathbf{a} - 3\mathbf{b} \times \mathbf{b}$$
$$= 2 \times \mathbf{0} + 6\mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{b} - 3 \times \mathbf{0} = 7\mathbf{a} \times \mathbf{b}.$$

30 c)

$$(5\mathbf{b} - 6\mathbf{a}) \times (6\mathbf{b} - 5\mathbf{a}) = 30\mathbf{b} \times \mathbf{b} - 25\mathbf{b} \times \mathbf{a} - 36\mathbf{a} \times \mathbf{b} + 30\mathbf{a} \times \mathbf{a}$$
$$= 30 \times \mathbf{0} + 25\mathbf{a} \times \mathbf{b} - 36\mathbf{a} \times \mathbf{b} + 30 \times \mathbf{0}$$
$$= -11\mathbf{a} \times \mathbf{b}$$
$$= 11\mathbf{b} \times \mathbf{a}.$$

31 a)

$$\begin{aligned}(\mathbf{a}+\mathbf{b})\times(\mathbf{a}-\mathbf{c}) &= \mathbf{a}\times\mathbf{a}-\mathbf{a}\times\mathbf{c}+\mathbf{b}\times\mathbf{a}-\mathbf{b}\times\mathbf{c}\\ &= \mathbf{0}-\mathbf{a}\times\mathbf{c}-\mathbf{a}\times\mathbf{b}-\mathbf{b}\times\mathbf{c}\\ &= -\mathbf{a}\times\mathbf{c}-\mathbf{a}\times\mathbf{b}-\mathbf{b}\times\mathbf{c}.\end{aligned}$$

31 b)

$$\begin{aligned}(\mathbf{c}-\mathbf{b})\times(\mathbf{b}-\mathbf{a}) &= \mathbf{c}\times\mathbf{b}-\mathbf{c}\times\mathbf{a}-\mathbf{b}\times\mathbf{b}+\mathbf{b}\times\mathbf{a}\\ &= -\mathbf{b}\times\mathbf{c}+\mathbf{a}\times\mathbf{c}-\mathbf{0}-\mathbf{a}\times\mathbf{b}\\ &= -\mathbf{b}\times\mathbf{c}+\mathbf{a}\times\mathbf{c}-\mathbf{a}\times\mathbf{b}.\end{aligned}$$

31 c)

$$\begin{aligned}&(\mathbf{a}-\mathbf{b}+\mathbf{c})\times(\mathbf{b}-\mathbf{a}+\mathbf{c})\\ &= \mathbf{a}\times\mathbf{b}-\mathbf{a}\times\mathbf{a}+\mathbf{a}\times\mathbf{c}-\mathbf{b}\times\mathbf{b}+\mathbf{b}\times\mathbf{a}-\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{b}-\mathbf{c}\times\mathbf{a}+\mathbf{c}\times\mathbf{c}\\ &= \mathbf{a}\times\mathbf{b}-\mathbf{0}+\mathbf{a}\times\mathbf{c}-\mathbf{0}-\mathbf{a}\times\mathbf{b}-\mathbf{b}\times\mathbf{c}-\mathbf{b}\times\mathbf{c}+\mathbf{a}\times\mathbf{c}+\mathbf{0}\\ &= 2\mathbf{a}\times\mathbf{c}-2\mathbf{b}\times\mathbf{c}.\end{aligned}$$

31 d)

$$\begin{aligned}&(2\mathbf{a}+\mathbf{b}-3\mathbf{c})\times(4\mathbf{a}-3\mathbf{b}+\mathbf{c})\\ &= 8\mathbf{a}\times\mathbf{a}-6\mathbf{a}\times\mathbf{b}+2\mathbf{a}\times\mathbf{c}+4\mathbf{b}\times\mathbf{a}-3\mathbf{b}\times\mathbf{b}+\mathbf{b}\times\mathbf{c}-12\mathbf{c}\times\mathbf{a}+9\mathbf{c}\times\mathbf{b}-3\mathbf{c}\times\mathbf{c}\\ &= 8\times\mathbf{0}-6\mathbf{a}\times\mathbf{b}+2\mathbf{a}\times\mathbf{c}-4\mathbf{a}\times\mathbf{b}-3\times\mathbf{0}+\mathbf{b}\times\mathbf{c}+12\mathbf{a}\times\mathbf{c}-9\mathbf{b}\times\mathbf{c}-3\times\mathbf{0}\\ &= -10\mathbf{a}\times\mathbf{b}+14\mathbf{a}\times\mathbf{c}-8\mathbf{b}\times\mathbf{c}.\end{aligned}$$

32)

$$\begin{aligned}(2\mathbf{a}-\mathbf{b})\times(6\mathbf{a}-5\mathbf{b}) = \mathbf{0} &\Rightarrow 12\mathbf{a}\times\mathbf{a}-10\mathbf{a}\times\mathbf{b}-6\mathbf{b}\times\mathbf{a}+5\mathbf{b}\times\mathbf{b}\\ &= \mathbf{0}\\ \Rightarrow 12\times\mathbf{0}-10\mathbf{a}\times\mathbf{b}+6\mathbf{a}\times\mathbf{b}+5\times\mathbf{0} = \mathbf{0} &\Rightarrow -4\mathbf{a}\times\mathbf{b}\\ &= \mathbf{0}\Rightarrow\mathbf{a}\times\mathbf{b}\\ &= \mathbf{0}.\end{aligned}$$

Therefore $\mathbf{a}$ and $\mathbf{b}$ are parallel.

33 a)

$$\begin{aligned}(\mathbf{a}+2\mathbf{b})\times(\mathbf{a}-\mathbf{c}) &= \mathbf{a}\times\mathbf{a}-\mathbf{a}\times\mathbf{c}+2\mathbf{b}\times\mathbf{a}-2\mathbf{b}\times\mathbf{c}\\ &= \mathbf{0}-\mathbf{a}\times\mathbf{c}-2\mathbf{a}\times\mathbf{b}-2\mathbf{b}\times\mathbf{c}\\ &= -\mathbf{a}\times\mathbf{c}-2\mathbf{a}\times\mathbf{b}-2\mathbf{b}\times\mathbf{c}\\ &= -\mathbf{j}-2\mathbf{k}-2\mathbf{i}\\ &= -2\mathbf{i}-\mathbf{j}-2\mathbf{k}.\end{aligned}$$

33 b)

$$\begin{aligned}(2\mathbf{a}+\mathbf{b}+\mathbf{c})\times(\mathbf{b}+\mathbf{c}) &= 2\mathbf{a}\times\mathbf{b}+2\mathbf{a}\times\mathbf{c}+\mathbf{b}\times\mathbf{b}+\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{b}+\mathbf{c}\times\mathbf{c}\\ &= 2\mathbf{a}\times\mathbf{b}+2\mathbf{a}\times\mathbf{c}+\mathbf{0}+\mathbf{b}\times\mathbf{c}-\mathbf{b}\times\mathbf{c}+\mathbf{0}\\ &= 2\mathbf{a}\times\mathbf{b}+2\mathbf{a}\times\mathbf{c}\\ &= 2\mathbf{k}+2\mathbf{j}\\ &= 2\mathbf{j}+2\mathbf{k}.\end{aligned}$$

33 c)

$$\begin{aligned}&(3\mathbf{a}-2\mathbf{b}-\mathbf{c})\times(\mathbf{c}-4\mathbf{b}-3\mathbf{a})\\ &= 3\mathbf{a}\times\mathbf{c}-12\mathbf{a}\times\mathbf{b}-9\mathbf{a}\times\mathbf{a}-2\mathbf{b}\times\mathbf{c}+8\mathbf{b}\times\mathbf{b}+6\mathbf{b}\times\mathbf{a}-\mathbf{c}\times\mathbf{c}+4\mathbf{c}\times\mathbf{b}+3\mathbf{c}\times\mathbf{a}\\ &= 3\mathbf{a}\times\mathbf{c}-12\mathbf{a}\times\mathbf{b}-9\times\mathbf{0}-2\mathbf{b}\times\mathbf{c}+8\times\mathbf{0}-6\mathbf{a}\times\mathbf{b}-\mathbf{0}-4\mathbf{b}\times\mathbf{c}-3\mathbf{a}\times\mathbf{c}\\ &= -18\mathbf{a}\times\mathbf{b}-6\mathbf{b}\times\mathbf{c}\\ &= -18\mathbf{k}-6\mathbf{i}\\ &= -6\mathbf{i}-18\mathbf{k}.\end{aligned}$$

34)

$$\begin{aligned}&(2\mathbf{a}-3\mathbf{b}+8\mathbf{c})\times(7\mathbf{a}+6\mathbf{b}-16\mathbf{c})\\ &= 14\mathbf{a}\times\mathbf{a}+12\mathbf{a}\times\mathbf{b}-32\mathbf{a}\times\mathbf{c}-21\mathbf{b}\times\mathbf{a}-18\mathbf{b}\times\mathbf{b}+48\mathbf{b}\times\mathbf{c}+56\mathbf{c}\times\mathbf{a}+48\mathbf{c}\times\mathbf{b}-128\mathbf{c}\times\mathbf{c}\\ &= 14\times\mathbf{0}+12\mathbf{a}\times\mathbf{b}-32\mathbf{a}\times\mathbf{c}+21\mathbf{a}\times\mathbf{b}-18\times\mathbf{0}+48\mathbf{b}\times\mathbf{c}-56\mathbf{a}\times\mathbf{c}-48\mathbf{b}\times\mathbf{c}-128\times\mathbf{0}\\ &= 33\mathbf{a}\times\mathbf{b}-88\mathbf{a}\times\mathbf{c}\\ &= 33\times3\mathbf{i}-88\times4\mathbf{j}\\ &= 99\mathbf{i}-352\mathbf{j}.\end{aligned}$$

35)

$$\begin{aligned}&(\mathbf{c}-\mathbf{b}-4\mathbf{a})\times(4\mathbf{a}-6\mathbf{b}+9\mathbf{c})\\ &= 4\mathbf{c}\times\mathbf{a}-6\mathbf{c}\times\mathbf{b}+9\mathbf{c}\times\mathbf{c}-4\mathbf{b}\times\mathbf{a}+6\mathbf{b}\times\mathbf{b}-9\mathbf{b}\times\mathbf{c}-16\mathbf{a}\times\mathbf{a}+24\mathbf{a}\times\mathbf{b}-36\mathbf{a}\times\mathbf{c}\\ &= -4\mathbf{a}\times\mathbf{c}+6\mathbf{b}\times\mathbf{c}+9\times\mathbf{0}+4\mathbf{a}\times\mathbf{b}+6\times\mathbf{0}-9\mathbf{b}\times\mathbf{c}-16\times\mathbf{0}+24\mathbf{a}\times\mathbf{b}-36\mathbf{a}\times\mathbf{c}\\ &= 28\mathbf{a}\times\mathbf{b}-3\mathbf{b}\times\mathbf{c}-40\mathbf{a}\times\mathbf{c}\\ &= 28\times2\mathbf{k}-3\times3\mathbf{j}-40\times5\mathbf{i}\\ &= -200\mathbf{i}-9\mathbf{j}+56\mathbf{k}.\end{aligned}$$

36)

$$(-2\mathbf{a}-6\mathbf{b}+3\mathbf{c})\times5\mathbf{a} = -10\mathbf{a}\times\mathbf{a}-30\mathbf{b}\times\mathbf{a}+15\mathbf{c}\times\mathbf{a}$$

$$
\begin{aligned}
&= -10 \times \mathbf{0} + 30\mathbf{a} \times \mathbf{b} - 15\mathbf{a} \times \mathbf{c} \\
&= 30 \begin{pmatrix} 4 \\ 1 \\ -3 \end{pmatrix} - 15 \begin{pmatrix} 8 \\ -3 \\ 2 \end{pmatrix} \\
&= \begin{pmatrix} 0 \\ 75 \\ -120 \end{pmatrix}.
\end{aligned}
$$

37)

$$
\begin{aligned}
&(3\mathbf{a} - \mathbf{b} + 2\mathbf{c}) \times (3\mathbf{a} - 2\mathbf{b} - 5\mathbf{c}) \\
&= 9\mathbf{a} \times \mathbf{a} - 6\mathbf{a} \times \mathbf{b} - 15\mathbf{a} \times \mathbf{c} - 3\mathbf{b} \times \mathbf{a} + 2\mathbf{b} \times \mathbf{b} + 5\mathbf{b} \times \mathbf{c} + 6\mathbf{c} \times \mathbf{a} - 4\mathbf{c} \times \mathbf{b} - 10\mathbf{c} \times \mathbf{c} \\
&= 9 \times \mathbf{0} - 6\mathbf{a} \times \mathbf{b} - 15\mathbf{a} \times \mathbf{c} + 3\mathbf{a} \times \mathbf{b} + 2 \times \mathbf{0} + 5\mathbf{b} \times \mathbf{c} - 6\mathbf{a} \times \mathbf{c} + 4\mathbf{b} \times \mathbf{c} - 10 \times \mathbf{0} \\
&= -6 \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} - 15 \begin{pmatrix} -2 \\ -1 \\ 4 \end{pmatrix} + 3 \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} + 5 \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} - 6 \begin{pmatrix} -2 \\ -1 \\ 4 \end{pmatrix} + 4 \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} \\
&= \begin{pmatrix} 24 \\ 30 \\ -72 \end{pmatrix}.
\end{aligned}
$$

38 a)

$$
\begin{aligned}
|(\mathbf{a} + \mathbf{b}) \times (2\mathbf{a} - \mathbf{b})| &= |2\mathbf{a} \times \mathbf{a} - \mathbf{a} \times \mathbf{b} + 2\mathbf{b} \times \mathbf{a} - \mathbf{b} \times \mathbf{b}| \\
&= |2 \times \mathbf{0} - \mathbf{a} \times \mathbf{b} - 2\mathbf{a} \times \mathbf{b} - \mathbf{0}| \\
&= |-3\mathbf{a} \times \mathbf{b}| \\
&= 3|\mathbf{a} \times \mathbf{b}| \\
&= 3|\mathbf{a}||\mathbf{b}| \sin 90^\circ \\
&= 3|\mathbf{a}||\mathbf{b}|.
\end{aligned}
$$

38 b)

$$
\begin{aligned}
|(3\mathbf{a} - 2\mathbf{b}) \times (4\mathbf{a} + \mathbf{b})| &= |12\mathbf{a} \times \mathbf{a} + 3\mathbf{a} \times \mathbf{b} - 8\mathbf{b} \times \mathbf{a} - 2\mathbf{b} \times \mathbf{b}| \\
&= |12 \times \mathbf{0} + 3\mathbf{a} \times \mathbf{b} + 8\mathbf{a} \times \mathbf{b} - 2 \times \mathbf{0}| \\
&= |11\mathbf{a} \times \mathbf{b}| \\
&= 11|\mathbf{a} \times \mathbf{b}| \\
&= 11|\mathbf{a}||\mathbf{b}| \sin 90^\circ \\
&= 11|\mathbf{a}||\mathbf{b}|.
\end{aligned}
$$

38 c)

$$
|(6\mathbf{b} - \mathbf{a}) \times (9\mathbf{a} - \mathbf{b})| = |54\mathbf{b} \times \mathbf{a} - 6\mathbf{b} \times \mathbf{b} - 9\mathbf{a} \times \mathbf{a} + \mathbf{a} \times \mathbf{b}|
$$

$$
\begin{aligned}
&= |-54\mathbf{a}\times\mathbf{b} - 6\times\mathbf{0} - 9\times\mathbf{0} + \mathbf{a}\times\mathbf{b}| \\
&= |-53\mathbf{a}\times\mathbf{b}| \\
&= 53|\mathbf{a}\times\mathbf{b}| \\
&= 53|\mathbf{a}||\mathbf{b}|\sin 90^\circ \\
&= 53|\mathbf{a}||\mathbf{b}|.
\end{aligned}
$$

39)

$$\mathbf{a}\times\mathbf{b} = |\mathbf{a}||\mathbf{b}|\sin\theta\,\hat{\mathbf{n}}.$$

If $\theta = 0^\circ$ then $\sin 0^\circ = 0$. If $\theta = 180^\circ$ then $\sin 180^\circ = 0$. If $\mathbf{a}$ and $\mathbf{b}$ are parallel, then the angle between them will be zero, hence $\mathbf{a}\times\mathbf{b} = \mathbf{0}$.

40)

$$
\begin{aligned}
&(\mathbf{a}+\mathbf{b})\times(\mathbf{a}-\mathbf{b}) = \mathbf{0} \\
\Rightarrow\ &\mathbf{a}\times\mathbf{a} - \mathbf{a}\times\mathbf{b} + \mathbf{b}\times\mathbf{a} - \mathbf{b}\times\mathbf{b} = \mathbf{0} \\
\Rightarrow\ &\mathbf{0} - \mathbf{a}\times\mathbf{b} - \mathbf{a}\times\mathbf{b} - \mathbf{0} = \mathbf{0} \Rightarrow -2\mathbf{a}\times\mathbf{b} \\
&\qquad\qquad = \mathbf{0} \Rightarrow \mathbf{a}\times\mathbf{b} \\
&\qquad\qquad = \mathbf{0}.
\end{aligned}
$$

So $\mathbf{a}$ and $\mathbf{b}$ are parallel, so the angle between the vectors is either 0° or 180°.

41)

$$
\begin{aligned}
\mathbf{a}\times\mathbf{b} = \mathbf{b}\times\mathbf{c} \Rightarrow \mathbf{a}\times\mathbf{b} - \mathbf{b}\times\mathbf{c} \\
= \mathbf{0} \Rightarrow \mathbf{a}\times\mathbf{b} + \mathbf{c}\times\mathbf{b} \\
= \mathbf{0} \Rightarrow (\mathbf{a}+\mathbf{c})\times\mathbf{b} \\
= 0.
\end{aligned}
$$

So $(\mathbf{a}+\mathbf{c})$ must be parallel to $\mathbf{b}$, hence $\mathbf{a}+\mathbf{c} = k\mathbf{b}$.

42)

$$
\begin{aligned}
|\mathbf{a}\times\mathbf{b}| &= |\mathbf{a}||\mathbf{b}|\sin\theta \Rightarrow 7\sqrt{2} \\
&= 4\times 3\sqrt{2}\sin\theta \Rightarrow \sin\theta \\
&= \frac{7}{12}. \\
\therefore \cos\theta &= \frac{\sqrt{12^2 - 7^2}}{12} = \frac{\sqrt{95}}{12}. \\
\mathbf{a}\cdot\mathbf{b} &= |\mathbf{a}||\mathbf{b}|\cos\theta \Rightarrow \mathbf{a}\cdot\mathbf{b} \\
&= 4\times 3\sqrt{2}\times\frac{\sqrt{95}}{12} \\
&= \sqrt{190}.
\end{aligned}
$$

43)

$$\begin{aligned}\mathbf{a}\cdot\mathbf{b} &= |\mathbf{a}||\mathbf{b}|\cos\theta \Rightarrow 10\sqrt{66}\\ &= 25\times 2\sqrt{3}\cos\theta \Rightarrow \cos\theta\\ &= \frac{\sqrt{22}}{5}.\end{aligned}$$

$$\therefore \sin\theta = \frac{\sqrt{5^2-(\sqrt{22})^2}}{5} = \frac{\sqrt{3}}{5}.$$

$$\begin{aligned}|\mathbf{a}\times\mathbf{b}| &= |\mathbf{a}||\mathbf{b}|\sin\theta \Rightarrow |\mathbf{a}\times\mathbf{b}|\\ &= 25\times 2\sqrt{3}\times\frac{\sqrt{3}}{5}\\ &= 30.\end{aligned}$$

44)

$$\begin{aligned}|\mathbf{a}\times\mathbf{b}| &= |\mathbf{a}||\mathbf{b}|\sin\theta \Rightarrow p\\ &= ab\sin\theta \Rightarrow \sin\theta\\ &= \frac{p}{ab}.\end{aligned}$$

$$\begin{aligned}\mathbf{a}\cdot\mathbf{b} &= |\mathbf{a}||\mathbf{b}|\cos\theta \Rightarrow q\\ &= ab\cos\theta \Rightarrow \cos\theta\\ &= \frac{q}{ab}.\end{aligned}$$

$$\begin{aligned}\sin^2\theta+\cos^2\theta = 1 &\Rightarrow (\frac{p}{ab})^2+(\frac{q}{ab})^2\\ &= 1 \Rightarrow \frac{p^2}{a^2b^2}+\frac{q^2}{a^2b^2}\\ &= 1 \Rightarrow p^2+q^2\\ &= a^2b^2.\end{aligned}$$

45 a)

$$\mathbf{b}\times\mathbf{c} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 2 & 3 & -1\\ 2 & -1 & 1\end{vmatrix}$$

$$\begin{aligned}&= (3\times 1-(-1)\times(-1))\mathbf{i}-(2\times 1-(-1)\times 2)\mathbf{j}+(2\times(-1)-3\times 2)\mathbf{k}\\ &= 2\mathbf{i}-4\mathbf{j}-8\mathbf{k}.\end{aligned}$$

$$\begin{aligned}\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) &= \begin{pmatrix}3\\-2\\1\end{pmatrix}\cdot\begin{pmatrix}2\\-4\\-8\end{pmatrix}\\ &= 3\times 2+(-2)\times(-4)+1\times(-8)\\ &= 6+8-8\\ &= 6.\end{aligned}$$

45 b)

$$\mathbf{b}\times\mathbf{c}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\3&-3&1\\4&1&4\end{vmatrix}$$
$$=((-3)\times 4-1\times 1)\mathbf{i}-(3\times 4-1\times 4)\mathbf{j}+(3\times 1-(-3)\times 4)\mathbf{k}$$
$$=-13\mathbf{i}-8\mathbf{j}+15\mathbf{k}.$$
$$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})=\begin{pmatrix}4\\16\\12\end{pmatrix}\cdot\begin{pmatrix}-13\\-8\\15\end{pmatrix}$$
$$=4\times(-13)+16\times(-8)+12\times 15$$
$$=-52-128+180$$
$$=0.$$

46)

$\mathbf{b}\times\mathbf{c}$ finds a vector perpendicular to both $\mathbf{b}$ and $\mathbf{c}$. If the scalar product between $\mathbf{a}$ and this perpendicular vector is zero, then $\mathbf{a}$ is also perpendicular to this vector. If $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are all perpendicular to $\mathbf{b}\times\mathbf{c}$, then $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ must be coplanar.

47)

$$\mathbf{a}\times\mathbf{b}=(a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k})\times(b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k})$$
$$=a_1b_1\mathbf{i}\times\mathbf{i}+a_1b_2\mathbf{i}\times\mathbf{j}+a_1b_3\mathbf{i}\times\mathbf{k}+a_2b_1\mathbf{j}\times\mathbf{i}+a_2b_2\mathbf{j}\times\mathbf{j}+a_2b_3\mathbf{j}\times\mathbf{k}+a_3b_1\mathbf{k}\times\mathbf{i}+a_3b_2\mathbf{k}\times\mathbf{j}+a_3b_3\mathbf{k}\times\mathbf{k}$$
$$=a_1b_1\times\mathbf{0}+a_1b_2\mathbf{i}\times\mathbf{j}+a_1b_3\mathbf{i}\times\mathbf{k}-a_2b_1\mathbf{i}\times\mathbf{j}+a_2b_2\times\mathbf{0}+a_2b_3\mathbf{j}\times\mathbf{k}-a_3b_1\mathbf{i}\times\mathbf{k}-a_3b_2\mathbf{j}\times\mathbf{k}+a_3b_3\times\mathbf{0}$$
$$=a_1b_2\mathbf{k}-a_1b_3\mathbf{j}-a_2b_1\mathbf{k}+a_2b_3\mathbf{i}+a_3b_1\mathbf{j}-a_3b_2\mathbf{i}$$
$$=(a_2b_3-a_3b_2)\mathbf{i}+(a_3b_1-a_1b_3)\mathbf{j}+(a_1b_2-a_2b_1)\mathbf{k}$$
$$=(a_2b_3-a_3b_2)\mathbf{i}-(a_1b_3-a_3b_1)\mathbf{j}+(a_1b_2-a_2b_1)\mathbf{k}.$$

48)

$$\mathbf{a}\times\mathbf{b}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\a_1&a_2&a_3\\b_1&b_2&b_3\end{vmatrix}$$
$$=(a_2b_3-a_3b_2)\mathbf{i}-(a_1b_3-a_3b_1)\mathbf{j}+(a_1b_2-a_2b_1)\mathbf{k}.$$
$$\mathbf{b}\times\mathbf{a}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\b_1&b_2&b_3\\a_1&a_2&a_3\end{vmatrix}$$
$$=(a_3b_2-a_2b_3)\mathbf{i}-(a_3b_1-a_1b_3)\mathbf{j}+(a_2b_1-a_1b_2)\mathbf{k}.$$

Then $-\mathbf{b}\times\mathbf{a}=-(a_3b_2-a_2b_3)\mathbf{i}+(a_3b_1-a_1b_3)\mathbf{j}-(a_2b_1-a_1b_2)\mathbf{k}$.

$$=(-a_3b_2+a_2b_3)\mathbf{i}-(-a_3b_1+a_1b_3)\mathbf{j}+(-a_2b_1+a_1b_2)\mathbf{k}$$
$$=(a_2b_3-a_3b_2)\mathbf{i}-(a_1b_3-a_3b_1)\mathbf{j}+(a_1b_2-a_2b_1)\mathbf{k}=\mathbf{a}\times\mathbf{b}.$$

Therefore $\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$.

49)

$$(\mathbf{a}+\mathbf{b})\times\mathbf{c} = \left(\begin{pmatrix}a_1\\a_2\\a_3\end{pmatrix}+\begin{pmatrix}b_1\\b_2\\b_3\end{pmatrix}\right)\times\begin{pmatrix}c_1\\c_2\\c_3\end{pmatrix}$$

$$= \begin{pmatrix}a_1+b_1\\a_2+b_2\\a_3+b_3\end{pmatrix}\times\begin{pmatrix}c_1\\c_2\\c_3\end{pmatrix}$$

$$= \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ a_1+b_1 & a_2+b_2 & a_3+b_3\\ c_1 & c_2 & c_3\end{vmatrix}$$

$$= ((a_2+b_2)c_3-(a_3+b_3)c_2)\mathbf{i}-((a_1+b_1)c_3-(a_3+b_3)c_1)\mathbf{j}+((a_1+b_1)c_2-(a_2+b_2)c_2)\mathbf{k}$$

$$= (a_2c_3+b_2c_3-a_3c_2-b_3c_2)\mathbf{i}-(a_1c_3+b_1c_3-a_3c_1-b_3c_1)\mathbf{j}+(a_1c_2+b_1c_2-a_2c_2-b_2c_2)\mathbf{k}$$

$$= (a_2c_3-a_3c_2+b_2c_3-b_3c_2)\mathbf{i}-(a_1c_3-a_3c_1+b_1c_3-b_3c_1)\mathbf{j}+(a_1c_2-a_2c_2+b_1c_2-b_2c_2)\mathbf{k}$$

$$= (a_2c_3-a_3c_2)\mathbf{i}-(a_1c_3-a_3c_1)\mathbf{j}+(a_1c_2-a_2c_2)\mathbf{k}+(b_2c_3-b_3c_2)\mathbf{i}-(b_1c_3-b_3c_1)\mathbf{j}+(b_1c_2-b_2c_2)\mathbf{k}$$

$$= \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ a_1 & a_2 & a_3\\ c_1 & c_2 & c_3\end{vmatrix}+\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix}$$

$$= \begin{pmatrix}a_1\\a_2\\a_3\end{pmatrix}\times\begin{pmatrix}c_1\\c_2\\c_3\end{pmatrix}+\begin{pmatrix}b_1\\b_2\\b_3\end{pmatrix}\times\begin{pmatrix}c_1\\c_2\\c_3\end{pmatrix}$$

$$= \mathbf{a}\times\mathbf{c}+\mathbf{b}\times\mathbf{c}.$$